

4 What is yield curve? What type of yield curve you interest in? (3 points)

A yield curve is graph showing the relationship between bond yields and time to maturity for bonds with the same credit quality usually US Treasury bonds.

3 main types of yield curves:

1. Normal (upward sloping) Yield curve

- Long term interest rate $>$ short-term rate
- signal for economic growth / expected inflation

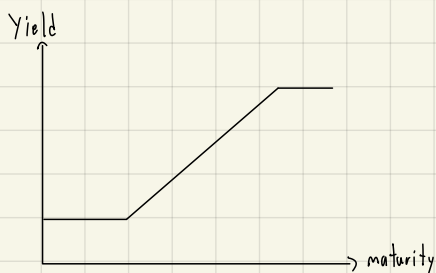
2. Flat Yield Curve

- short term $\approx$ long term yields close together
- Often signal economic slowdown

3. Inverted Yield Curve

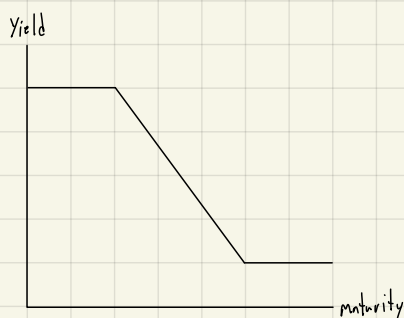
- short-term rate $>$ long term rate
- predictor of recession

5 Draw steep yield curve. What will happen on economy (4 points)



1. Economic expansion / recovery investors expect stronger economic growth in future
2. Higher inflation expectations investor demand higher long-term yields because inflation rise.
3. Future interest rate central bank may rise short-term interest rate to control inflation
4. More borrowing and investment activity business and consumer become more confident tend to spend / invest more.

6 Draw inverted yield curve. What will happen on economy (4 points)



1. Possible recession inverted yield curves often appear before recessions.
2. Economic slowdown investors expect weaker economic growth in future
3. Lower future interest rates market expect central bank to cut interest rate later
4. Lower inflation expectations investors expect inflation and demand to decrease

1. Explain thoroughly why interest rate and yield curve are important to the economy. What interests are you interested in? Do you prefer high or low level of interest rate? (10 pts)

Interest rate and the yield curve are very important to economy because they affect borrowing, spending, investment, inflation, and economic growth. Interest rates represent the cost of borrowing money and the return from saving or investing. When interest rate change, households, firm, bank and investor all adjust their behavior.

Low interest rate encourage consumer and business to borrow and spend/invest more. This can lead to increase economic growth and employment. However if interest rate are low inflation may rise due to demand strong,

High interest rate reduce borrowing / spending because loan expensive. Central bank may increase interest rate due to inflation to control. Moreover, high rate can slow inflation but slow economic growth and increase unemployment.

The yield curve important because show relationship bond yield and maturity help understand expectation future interest rate and the economy. Upward slope economic expansion while inverted yield curve signal future recession.

I am interested in long-term interest rate because they strongly affect investment, mortgages, bond market and future expectation.

I prefer low and stable interest rate because they support investment and business and economic growth. However, rate should not be low because low rate can lead to inflation.

2. The economy is going into recession period so the central bank implements an expansionary monetary policy and the government increases expenditure. What should happen to nominal and real interest rates? Which components of interest rate should change? Explain thoroughly by applying theories and economic letters. (10 pts)

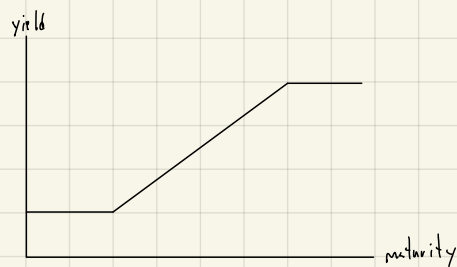
When the economy enters a recession, the central bank uses expansionary monetary policy and government increase spending to stimulate aggregate demand and economic growth.

Under expansionary monetary policy, the central bank low short-term rate policy and increases money supply. Therefore, nominal and real interest rate should decrease to encourage borrowing investment and consumption.

Use Fisher Equation $i = r + \pi^e$ Real interest rate should decrease to stimulate invest/spend. Expected Inflation may increase because expansionary policy may raise future inflation expectations. Risk premium may change depend on unstable financial markets.

A yield curve theory, expansionary monetary policy lower short-term yield than long term yield, yield curve may steeper during recovery. Steep yield curve signal expectation future economic growth and inflation.

3. If the economy is going into boom period, draw and explain the shape of yield curve. Furthermore, if the level of inflation is rising and the central bank implements monetary policy, what should happen to the adjustment of yield curve? Draw the movement of yield curve and explain thoroughly by applying theories and economic letters. (10 pts)



During boom period, yield curve is upward sloping or steep because investors expect stronger economic growth high inflation and high future interest rate. Long-term yields higher than short term yield because investors demand extra return for expected inflation and future rate increase

If inflation keep rising central bank will contractionary monetary policy by raising short-term interest rate to control inflation



monetary policy strongly affects the slope of the yield curve. When the central bank raise short-term interest rate, short-term yields increase faster than long term yield causing curve to flatten. Moreover continue contractionary monetary policy can lead to invert and signal future economic slowdown

4. What should cause long run interest rate to change? Explain thoroughly by applying theories and economic letters. (10 pts)

Long run interest rates change because change inflation expectations, real interest rate monetary policy expectation economic growth, and risk premium

Expectation Theory, long term interest rate reflect expectation short term interest rate. If investor expect central bank will raise short term interest rate long term yield will increase

Fisher Equation $i = r + \pi^e$

long run nominal interest rate increase when Expected inflation rise and real interest rate rise

- Expected inflation is important to drive long term yield if investor expect higher future inflation they demand higher yield to protect purchasing power
- Long run interest rate are effect economic growth expectation. When strong economic growth investor will expect high inflation and increase in policy rate causing long term yield rise.
- Risk premium are important to drive yield investor require extra compensation for long-term bond because riskier long term yield must rise

5. Based on the following financial ratios, what do you think about the financial condition of this firm? (10 pts)

Year	2019	2020	2021
1. Current Ratio	0.52	0.78	0.60
2. Debt Ratio	0.80	0.92	0.75
3. Total Asset Turnover	1.43	1.58	1.67
4. ROE	0.45	0.58	0.60
5. PE Ratio	1.35	1.65	1.55

1. Current Ratio : Good > 1.0 Strong > 2.0 Weak < 1.0

The current ratio below 1 all years mean firm may not have enough current asset to cover short-term liabilities

However it improve in 2020 but decrease in 2021 this show weak liquidity may have problem to pay short-term debt.

2. Debt Ratio : stable < 0.5 High leverage > 0.5

The debt is very high all years. This mean most asset finance by debt.

Ratio increase in 2020 to 0.92 or 92% of asset were finance by liability very risky

but 2021 is improve but still high leverage. This create high finance risk and high pay interest.

3. Total Asset Turnover Higher ratio = better efficiency

The ratio improved every year, which means company use asset efficient to generate revenue

This positive sign for operational performance

4. ROE : good $> 10-15\%$ Higher = better profitability

ROE increase steadily which mean firm generate higher return for investor

However ROE can raise by leverage increase financial risk

5. P/E Ratio High P/E = growth expectation Low P/E = weak growth expectation

The P/E ratio relative low may have limit growth expectation or concern about firm risk and financial condition

Increase in 2020 reflect investor confident but decline in 2021 suggest unstable.

Overall financial condition

Company's strong

- improve efficiency asset return

- improving profitable high ROE

But

- weak liquidity (low current ratio)

- High leverage (High debt ratio)

The firm operationally efficient but finance risky because relies on debt finance

The company should improve liquidity and reduce debt to strong long-term finance stable.